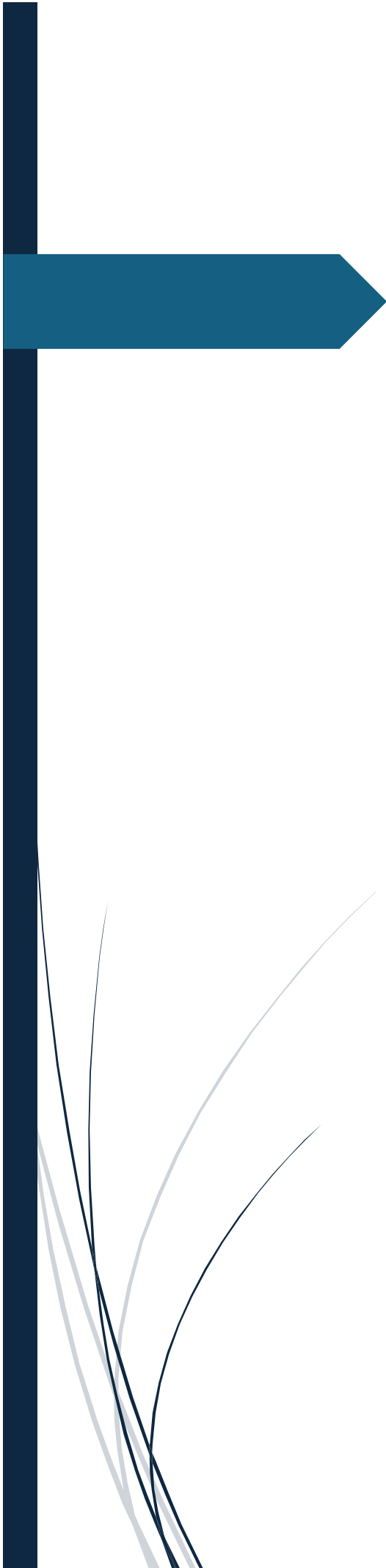




Statistical Machine Learning for Risk and Actuarial Applications Assignment: Hospitalised COVID- 19 Patients in Brazil

Assignment 2

Tanvir Randhawa

Executive Summary

We used machine learning to analyse the likelihood of COVID-19 death based on patients' characteristics in Brazil. In part 1 we developed an inferential interpretable model leveraging all available information. In part 1, we decided to use an Elastic Net model which incorporated regularisation and had high interpretability along with excellent performance and minimal error. In part 2 we fitted several models to find the best predictive model with limited data and decided to settle on a logistic model with interaction and non-linear terms to address the trends in age we found in both parts 1 and 2. From both parts, we found age to be an important predictor. Through this, we can guide insurers in risk stratification and aid hospitals in identifying high-risk patients.

Decision Criteria – For Part 1 and 2

We used the direct comparison method of 10-fold cross-validation to tune hyper-parameters. We used 10 folds due to its computational efficiency with our data set and reliable estimation. We split the data into training and testing sets in an 8:2 ratio. These sets were separated to prevent data leakage which would lead to unreasonably good predictions. We also removed high-leverage points since we realised there was a significant difference in results. For instance, part 1's final model 1 had a cross-validation misclassification rate of 0.04 lower without high leverage points.

To assess the fit of our models on training data, we used the cross-validation misclassification rate. This rate is dependent on different thresholds and uses different folds of the training data to estimate the test error whilst in the training phase of the model allowing for iterations. Models with lower cross-validation misclassification rates generally make more accurate predictions. This is important in our case since we want to minimise incorrect classifications for those who died from COVID-19 based on characteristics to maximise efficiency in dealing with high-risk patients. We used the F1 metric as it balances precision and recall, thus providing a comprehensive performance measure given false positives and false negatives have significant consequences. This is an ideal metric since it minimises overdiagnosis and missed diagnoses to ensure high-risk patients are identified and treated but also maintains efficient resource allocation by minimising the misallocation of resources which is important in part 2. Its effectiveness in handling imbalanced datasets and evaluating critical errors makes it valuable for these high-stakes applications. Since F1 scores may be similar between models, we used AUC to validate results since it provides a comprehensive evaluation of a model's performance across thresholds, is robust to class imbalances, and offers clear interpretability.

1.0 – Part 1

To fit an interpretable model for COVID-19 death likelihood, we understood that regularisation would be beneficial, as Géron (2022) claims that it is almost always preferable to have some regularisation. There are benefits since there is no need to manually drop variables since the regularisation process shrinks the coefficients and assigns higher coefficients to more statistically significant predictors, thus observers can easily interpret the model by differentiating between the direction and magnitude of the coefficient and eliminate all predictors with coefficient 0.

1.1 – Lasso Model

We fitted a Binomial GLM with the logit link and a Lasso l_1 penalty, for this and all future models we fitted we used a binomial model due to the binary nature of the response variable. The l_1 penalty addresses multicollinearity, thus stabilising the model by using the sum of absolute values of the parameters and pushing some coefficients to 0 to prevent overfitting. As Géron also suggested, Lasso models are preferable when a relatively smaller number of predictors are significant. However, we do not believe this to be the case since COVID-19 is a complex condition that has various factors contributing to death. We found that the cross-validation misclassification rate was relatively high at 0.191377. With the optimal threshold for the predictions is 0.5, this model had a weak AUC value of 0.7181733 which suggests that the model is not as strong at differentiating between positive and negative cases. This is further supported by the F1 score of 0.7825408 which is less than we expected.

1.2 – Ridge and Elastic Net Model

We fitted a Binomial GLM with the logit link and a Ridge l_2 penalty. Ridge has benefits since the penalty addresses multicollinearity and does not shrink coefficients to 0, this is more useful when examining issues where the result has relatively less important predictors that may contribute minutely. This is slightly less interpretable since there is more analysis required to assess magnitude. When tuning our models we considered the bias-variance trade-off which is prominent in shrinkage models, causing the variance to decrease significantly due to less overfitting, however, this causes bias to increase. To limit this extreme when choosing the best penalty parameter, we realised the λ that minimises the binomial deviance would inflate bias hence we used the λ value which was one standard deviation away from the 'best', we used a grid of λ values between 10^{-10} and $10^{0.05}$, and applied this method to all regularisation models. The cross-validation misclassification rate for the Ridge model was 0.1808712 which is less than Lasso's. This shows improvements from restricting the l_1 penalty. Further, the F1 score improved to 0.7877779 and AUC to 0.784037.

We fitted an Elastic Net Model to test if limiting the l_1 penalty would be beneficial, we placed an equal shrinkage parameter on both penalties $\lambda = 0.00828$. We found that 10 of the 29 variables were shrunk to 0, and 3, close to 0 which allows for more interpretability since we can differentiate between predictors which have essentially no influence and those with minor influences. This model provided a lower cross-validation misclassification error of 0.1800661 which shows a decent improvement. This is furthered by the model's excellent performance with the high F1 score of 0.7925172, evincing the model has a well-balanced performance on unseen data whilst concurrently attaining high precision and recall. The higher AUC value of 0.7963935 means this model is still capable of differentiating cases robustly although not the priority. Thus, we chose the Elastic Net model.

1.3 – Spline and Polynomial Implementations

The residuals QQ plot in Figure 1 indicated that the model's linear assumption is insufficient since the residuals are not normally distributed, suggesting non-linear patterns in the data. To address this, we experimented with spline and polynomial terms.

We considered a polynomial implementation with the Elastic Net model with the non-binary predictor age. Upon testing with and without the polynomial age terms, we found that our model performed best with a degree 2 age term suggesting that it is a non-linear predictor, specifically quadratic. The cross-validation misclassification error is 0.1800541. This shows a slight improvement from the model without the polynomial implementation meaning more of the relationships are considered, hence the model can explain more of the results. Further, the F1 scores increased to 0.7925220, this shows that both recall and precision have improved. However, the AUC has largely remained consistent at 0.7963980 showing a weaker ability to differentiate between classes, and hence we explored spline implementations.

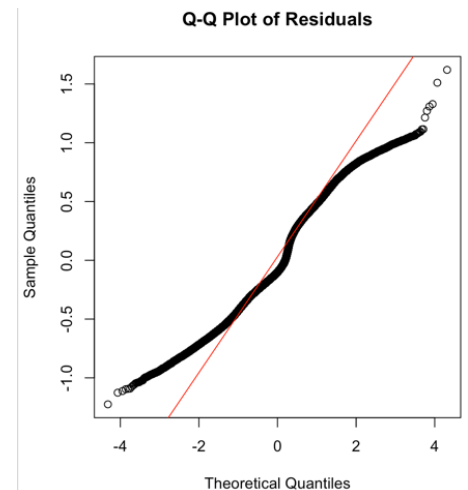


Figure 1 - Q-Q Plot of residuals

We found natural spline to perform better than with age having a degree of freedom of 2, this allowed for the inclusion of a knot which increased flexibility. The cross-validation misclassification error rate minutely decreased to 0.1800481 which shows that the model is more accurate and thus is performing better at distinguishing between classes. The F1 score contributes to this evaluation as it significantly improved to 0.7936190 and AUC to 0.7963995, this portrays that the model understands the underlying trend better as opposed to the noise. This could be because the spline introduced more adaptability to new data and can avoid some of the pitfalls of polynomials. This is preferred in these cases since there is a large amount of data.

Additionally, each segment of the spline can be interpreted independently, making it easier to interpret how the model behaves over different age ranges. This local flexibility with minimal overfitting with the regularisation reduces bias and provides a better representation of the underlying data trends which allows it to perform better on unseen data. By incorporating splines, we account for the non-linearity highlighted in the QQ plot.

1.4 – Final Model Justification and Exploration of Variables

Through the quantitative comparison of the decision criteria, binomial GLM with Elastic Net regularisation and a spline where age had a degree of freedom of two transcends all other models in all metrics. The regularisation allows our model to be parsimonious since it is simplified where only the most important predictors are considered. Further, this model has the lowest false negative rate of 0.26, this rate must be low to minimise missed cases which would mean those patients who require more attention do not receive this care which would inflate the mortality rate.

The benefits of both regularisation penalties prevented overfitting and multicollinearity. This lack of multicollinearity was further supported by the correlation values between all the coefficients that we generated in a correlation matrix which were all under 0.5. This is an easily interpretable model since those coefficients shrunk to 0 hold essentially no importance in this model and those very small have a slight but noteworthy contribution in the medical setting. By using all predictors, the model can leverage the full information. This is useful when there are many predictors, some of which have a small but statistically significant contribution.

We quantitatively compared the coefficients seen in Figure 3 and found that age which is modelled using natural splines has high positive coefficients indicating a significant relationship between age and the log odds of COVID-19 death and that the impact of age on mortality risk is not constant across ages. 0.6493 represents the log odds for ages before the knot and likely coincides with infant mortality. The log odds are higher after the knot emphasising that infant diagnosis with COVID-19 is a slightly lesser concern than elderly mortality. 0.7143 are the log odds for the ages after the knot,

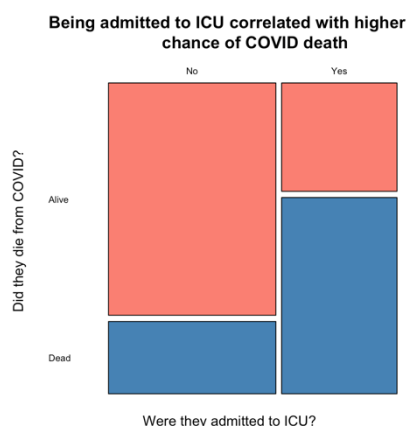


Figure 2 – Mosaic Plot of ICU Mortality

(0.0636), are immunocompromised (0.0703), or have neurological conditions (0.0548), are associated with increased log odds of death. Lower oxygen saturation (0.0340) has a slightly lower positive log odd coefficient, yet is a critical factor, as it relates to respiratory function which respiratory distress also has a positive log odds of 0.0211, a key concern in severe COVID-19. The number of comorbidities has a positive coefficient of 0.0308 and suggests that each additional comorbidity incrementally increases the log odds of death. The coefficient for cough is slightly negative (-0.0173), which suggests that cough alone, is less indicative of death. The impact of sex shows that males have a 0.0119 increase in log odds of death from COVID-19 and pneumopathy (0.0121) is minor but is still valuable for the risk assessment. The other predictors in Figure 3 have been shrunk to 0 evincing they have no notable impact on COVID-19 death log odds. Ultimately, age and ICU

this coefficient is the highest meaning log odds of death increase more after this knot. The ICU variable has a moderately positive coefficient of 0.3779, suggesting that being in the ICU decently increases the log odds of death, this indicates that ICU admission reflects severe illness. As seen in Figure 2, most ICU-admitted patients died due to COVID-19 and this proportion is significantly larger than those not admitted who had a higher likelihood of surviving.

Conversely, the relatively negative coefficient for vaccination, -0.0820 demonstrates its protective effect, reducing the risk of death among COVID patients. Certain pre-existing conditions highlight a patient's vulnerability, since those who have renal diseases (0.0636), are immunocompromised (0.0703), or have neurological conditions (0.0548), are associated with increased log odds of death. Lower oxygen saturation (0.0340) has a slightly lower positive log odd coefficient, yet is a critical factor, as it relates to respiratory function which respiratory distress also has a positive log odds of 0.0211, a key concern in severe COVID-19. The number of comorbidities has a positive coefficient of 0.0308 and suggests that each additional comorbidity incrementally increases the log odds of death. The coefficient for cough is slightly negative (-0.0173), which suggests that cough alone, is less indicative of death. The impact of sex shows that males have a 0.0119 increase in log odds of death from COVID-19 and pneumopathy (0.0121) is minor but is still valuable for the risk assessment. The other predictors in Figure 3 have been shrunk to 0 evincing they have no notable impact on COVID-19 death log odds. Ultimately, age and ICU

Predictor	Coefficient
ns(age, df = 2)2	0.714327971
ns(age, df = 2)1	0.649302142
icu	0.377880708
(Intercept)	-0.087165661
vaccine	-0.081993228
immuno	0.070331957
renal	0.063616718
neurological	0.054841682
oxygensat	0.033975508
diseaseCount	0.030816847
dyspnoea	0.025726779
respdistress	0.021060969
cough	-0.017349076
pneumopathy	0.012073967
asthma	-0.011856208
sex	0.011854875
cardio	-0.007252097
fever	0
sorethroat	0
diarrhea	0
vomit	0
hematologic	0
downsyn	0
diabetes	0
obesity	0
hepatic	0
icuStayDuration	0
hospStayDuration	0
symptomsCount	0

Figure 3 – Table of Coefficients – Log Odds

Understanding that age and ICU admission are major risk factors allows insurers to adjust premiums accordingly. Vaccination status can be incorporated into policies, potentially offering lower premiums or benefits to vaccinated individuals. Hospitals should also ensure to prioritise patients at high-risk ages specifically the elderly, they should also separate patients based on vaccination status. The identification of high-risk groups helps insurers to stratify risk and effectively manage claims.

1.5 – Limitations and Assumptions in Models

Although splines address non-linearity by fitting piecewise polynomial segments to predictors, specifically, age in our model, the GLM relies on an assumption of linearity for other predictors. Figure 1 suggests a non-normal distribution of residuals, and this approach may not have captured all potential non-linear relationships. Regularisation may introduce heteroscedasticity, violating the homoscedasticity assumption, by shrinking coefficients it reduces variance but affects the residuals' distribution. Error variability now depends on predicted values, increasing biased estimates. By constraining the coefficients, Elastic Net regression might not capture the variability. If the predictors have different scales, the shrinkage penalties may disproportionately affect certain coefficients, which can further lead to residuals with non-constant variance. It is assumed that all relevant predictors have been provided however, we understand that realistically, some confounding variables may not be included, and this can cause omitted variable bias, hindering the accuracy of results. We have found potential confounders to be smoking history and ethnicity as found by Bhaskaran, 2021. Further, the assumption of predictor independence might not hold in this scenario where covariates can be interrelated if we conduct a deeper analysis into causation rather than just correlation. It is assumed that there is no multicollinearity due to the implementation of the regularisation however, if there is still multicollinearity it can inflate standard errors and decrease estimate reliability. We should consider a tree-based model to address this linearity assumption, however, this can present challenges since they are prone to overfitting and bagging or boosting techniques can present high variance.

2.0 – Part 2

A predictive model is useful for insurers to gain a stronger understanding of links between patients' characteristics known upon hospital admission and their likelihood of death from COVID-19. The aim is to aid medical practitioners in providing correct services to patients sooner. Predictive models are concerned with the outcome and statistical significance of predictors rather than explaining results.

2.1 – Regularisation Models: Lasso and Ridge

Although the Ridge model had significant benefits in part 1 where more predictors were available, when fitting the model with limited predictors, we found that the cross-validation misclassification rates were higher, 0.2866013 for Lasso and 0.2917032 for Ridge. The performance is also lower for Lasso and Ridge with an F1 score of 0.61055 and 0.60998 respectively portraying a weaker precision

and recall, this was also supported by the weak differentiating ability with AUC values of 0.6938214 for Lasso and 0.688991. When constructing these models, we used the shrinkage coefficient which was one standard deviation away from the ‘best’ value to ensure that there was no significant variance in the results. We suspect regularisation oversimplified the models and added too much bias; thus the model cannot pick up on lesser significant trends that have some influence on COVID-19 mortality.

2.2 – KNN Model

We fitted a KNN model, however, it also performed poorly across all metrics providing an F1 score of 0.59081. This may be because KNN can be sensitive to noisy data points or outliers. The performance of KNN heavily depends on the choice of the number of neighbours, k . A small k can lead to overfitting, whilst a large k can lead to underfitting, through cross-validation with odd k values to avoid ties in the ‘voting’ system, we found a k of 5 to provide the best AUC and F1 score which was still low indicating a poor balance between precision and recall. This classification method is based on the ‘nearness’, and hence is less objective. Since the aim is not interpretability which KNN excels in, we opted against it as a predictive model.

2.6 – Tree Model – Random Forest and Single Decision Tree

We initially fitted a single decision tree (Appendix 9), and although it was easily interpretable, it had weak predictive performance compared to the ensemble method of random forests only providing an F1 score of 0.54542. This was likely due to overfitting, which trees are prone to.

To improve from this basic decision tree, we built a random forest. We found this to be a very computationally expensive process, hence we applied the model of 50 trees to get a gauge for the potential performance with our predictors and then also tried 100 trees which provided similar results. We also used 8 predictors in each tree which is different to the recommended rule of the square root of predictors, however, since we had a limited number of predictors, we opted for this method to observe more trends. we used cross-validation by finding the number of nodes which minimises the deviance. Random Forest was also a relatively weaker predictive model, only providing an F1 score of 0.59156 which although better than that of the single decision tree is still poor. The AUC of 0.70932 further emphasises the model’s inability to differentiate between cases and poor balance between recall and precision.

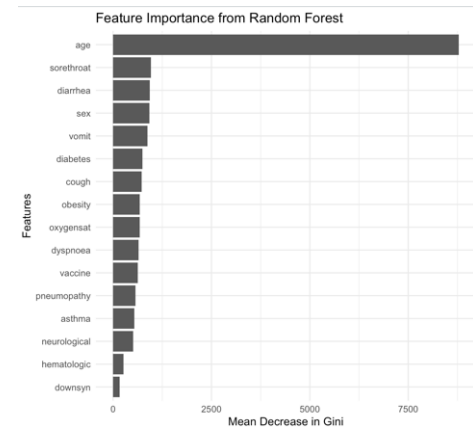


Figure 4 – Random Forest Importance Plot

A benefit of tree-based methods is that they capture non-linear trends and there is no need to tune the threshold, we can examine the node purity to determine where the most positive cases are. We generated a variable importance plot, although we decided against a random forest, we gained valuable insight into what the most important predictors were. This plot aligns with our findings for part 1 since age also has the most significant impact on COVID-19 death likelihood based on the Gini metric. This impurity metric shows that since there is a higher mean decrease in Gini for age, the predictor has the highest significance in partitioning the data into purer nodes. Conversely, down syndrome, hematology and neurologic conditions do not have as strong of importance suggesting these are less important to consider which is reflected by the coefficients in section 2.7. This plot supports our claims in 2.7 emphasising that hospitals should prioritise elderly patients.

2.3 – Logistic Model

In our logistic regression model, we included all predictors that were both continuous and binary, we found the logit link function to be effective as it provided a higher F1 score compared to probit, this could be because the Sigmoid function excels in binary situations and is more responsive to extreme results which improved its precision. To avoid overfitting, we included only statistically significant predictors, hence dropped "vomit" and "fever", which improved the model's parsimony and slightly increased the F1 score from 0.61069 to 0.61096. We decided to iterate this model further.

2.4 – Interaction Terms

In our analysis of COVID-19 mortality factors, we created various interaction terms to explore the combined effects of covariates to uncover complex relationships that individual predictors might miss. A correlation matrix (Appendix 1) guided our selection, as we investigated predictors with moderate correlations. We conducted trial and error with various interaction terms with the logistic regression. Notably, interactions between fever and vaccination improved the model’s performance, as indicated by enhanced F1 scores from 0.61182 to 0.61225. However, the interaction between age and vaccination, while statistically significant and beneficial to model performance, had a VIF of 32.87, exceeding the threshold of 10, indicating severe multicollinearity. This high VIF value suggested that the predictor was highly correlated with others, leading to potential instability in the model which is possibly why we attained misleadingly high F1 scores. We did not include this interaction term since a core component of a strong predictive model is stability which this hinders.

2.5 – Polynomial and Spline Implementations

We experimented with a polynomial term for age of degree 2 to model non-linear effects, but it risked overfitting. Hence, we explored splines of degrees of freedom 2-4 for age, we found the optimal degrees of freedom as 2 which aligns with section 1.3, this spline provided greater flexibility and better adaptation to the underlying trends. The spline terms were highly statistically significant and reduced overfitting, as seen by the improved F1 score of 0.00012 compared to the polynomial version.

To tune the threshold, we used cross-validation. We found the optimal threshold to be 0.3-0.31 for all models. Since it is quite low, it usually coincides with a high false positive rate. The FPR was found to be 62.07% for this final model, this is concerning as it shows a weaker model, however, in the medical setting it is safer to have a higher type 1 error than a type 2 error as discussed by Huecker and Shreffler, 2024. It means those patients will be more catered to and supported to aid recovery preventing them from being overlooked. This is consistent with our results since the false negative rate for this model is 25.93%. Although these are lower, they are concerning since time and resources must now be divided amongst more patients, some of whom do not require it.

2.7 – Final Model Justification

Our final model is a GLM with a natural spline with a degree of freedom 2 for age and an interaction term between fever and vaccine. This model has the highest F1 score of 0.61257 and the highest AUC value of 0.760561 which means that it is comparatively excellent at differentiating between cases whilst balancing the precision and recall better than other models. This emphasises that the model is most competent at predicting a patient's death likelihood upon hospital admission, although it is statistically weaker than the model in part 1 which leverages more information. We analysed VIF values which were all found to be between 1 and 1.9, hence multicollinearity is not an issue.

The spline terms for age have high coefficients. The ages before the knot have a higher likelihood of death from COVID-19 (4.506), this coincides with infants and young children since they supposedly do not have strong immune functions. Hence, very young patients are high risk and should be treated first, to prevent fatality. The log odds of death after this knot are only slightly less, with the log odds increasing by 3.508 without considering other factors. This underscores the heightened vulnerability of older patients. To address this, hospitals should admit very young and old patients to the ICU and address conditions sooner. It cannot be said if age causes a heightened risk of death, hence we assume a correlation. The slight positive log odds for sex (0.19) suggest that males have a higher risk, and the moderately negative coefficient for vaccination status (-0.47) demonstrates the slight protective effect of vaccines. This suggests that vaccinated patients should not be considered as high-risk without yet considering other factors. Although fever was not a statistically significant predictor, the combined effect of fever and vaccine is seen through the decreased log odds of death (-0.056), further exemplifying the benefits of vaccination. Symptoms such as cough (-0.24) and dyspnoea (0.40) show varying impacts on log odds, with dyspnoea indicating heightened illness and increased mortality risk. Symptoms of diarrhea, cough, and sore throat suggest decreased log odds of death to varying degrees as seen through the log odds in Figure-5. Comorbidities like diabetes (0.22), obesity (0.37), and hematologic conditions (0.29) substantially elevate the log odds of mortality, so hospitals should classify these patients as high-risk.

Predictor	Estimate
(Intercept)	-2.892971133
ns(age, df = 2)1	4.50590586
ns(age, df = 2)2	3.50789032
sex	0.187577998
vaccine	-0.472479095
cough	-0.237669879
sorethroat	-0.046613331
hematologic	0.292408266
downsyn	0.471380915
diarrhea	-0.115565322
dyspnoea	0.404996091
oxygensat	0.438155407
asthma	-0.177516091
diabetes	0.21659836
neurological	0.398340168
pneumopathy	0.312164788
obesity	0.372532269
fever_vaccine	-0.055909728

Figure 5 – Table of Coefficients – Log Odds

This analysis suggests that infants and older, male, unvaccinated patients with severe symptoms such as dyspnoea, low oxygen saturation and multiple comorbidities have the highest risk for COVID-19 hence. If patients have more of these qualities, they should be more prioritised by hospitals, and attain greater attention upon admission this should contribute to better risk management hence less fatalities.

2.9 – Limitations and Assumptions in Models

Again, there is the underlying assumption of linearity besides the inclusion of spline interactions the model's residuals are not normally distributed as seen in Appendix 12, as discussed in section 1.5 the accuracy of the results is hindered. There is also the problem of omitted variable bias since we suspect there are confounder variables that have not been considered. To improve the validity of this model, we as insurers should investigate how to attain more patient information upon hospital admission.

Conclusion

Through our analysis, we were able to identify key risk covariates behind COVID-19 death, especially age and factors that prompt ICU admission. Both final models had the strongest predictive performance and lowest cross-validation misclassification rates. The Elastic Net model's superior interpretability with COVID-19 death data can help us insurers adjust premiums, manage claims more effectively, and design targeted interventions. The parsimonious predictive model will allow hospitals to better stratify risk, and be resource-efficient, thus preventing high mortality rates.

Technical Appendix

Data Pre-Processing

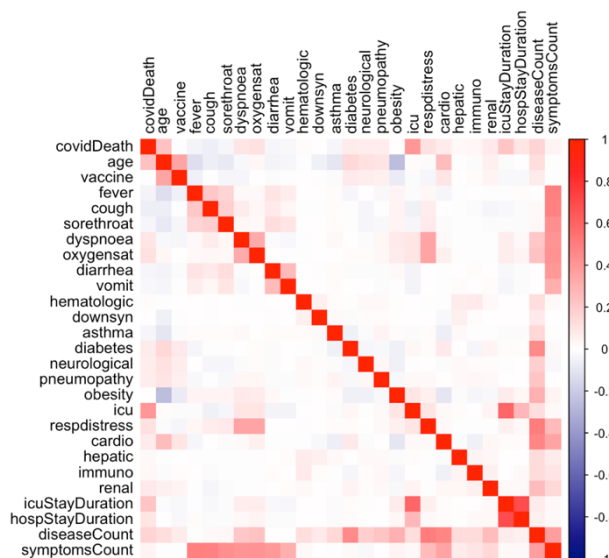
When checking for missing entries or duplicates, we found there to be no missing data nor duplicates.

Variables Omitted

Would be computationally exhausting if patient ID and dates were included, hence they were omitted, and intuitively they were not necessary

Variables Added

Added variables for Age group, ICU stay duration, Hospital stay duration, the cumulative number of symptoms, the cumulative number of pre-existing diseases. This could only be used for part 1.



Through the interpretation of this correlation matrix, it was clear that there were many variables that had a strong correlation with COVID-19 death, in both positive and negative directions. These included ICU, age, respiratory distress, oxygen saturation, dyspnoea, Hospital stay duration, diabetes, pneumopathy, neurological, and cough. We used this to assist us when we experimented with interaction terms since often times these terms have high correlation values.

Appendix 1

Part 1 – Note Most of the discussion is in the report – This is supplementary findings

Ridge Model

	Coefficient
icu	0.3856256686
(Intercept)	-0.2758181980
immuno	0.1163869871
renal	0.1083451856
hepatic	0.1072181803
neurological	0.0956104169
vaccine	-0.0952316862
downsyn	0.0652410602
respdistress	0.0560388380
pneumopathy	0.0542755176
oxygensat	0.0437412999
dyspnoea	0.0368025683
obesity	0.0327426255
diabetes	0.0316299281
cough	-0.0254730612
sex	0.0224953933
hematologic	0.0213322059
asthma	-0.0156522259
age	0.0074909723
cardio	0.0048713224
vomit	0.0045135236
diarrhoea	-0.0042466454
symptomsCount	-0.0033306099
fever	-0.0014546874
hospStayDuration	-0.0011085205
icuStayDuration	0.0003295835
diseaseCount	0.0002523197
sorethroat	-0.0002058307

This model was slightly hard to interpret as we were simply dealing with a lot of numbers very close to 0 and some were just negligible so the lasso penalty was necessary because the goal of this part of the assignment was interpretability. This is not the best model.

We used a lambda value of 0.0537 which was one standard deviation away from the mean.

ICU had a higher ratio here as well

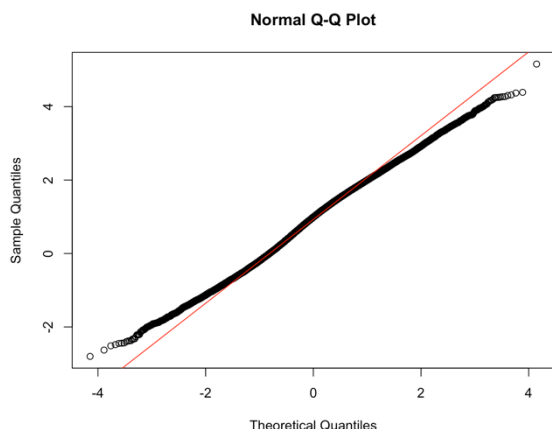
Appendix 2

Lasso Model

	Coefficient
(Intercept)	-3.41815820
icu	1.35440527
immuno	0.48094999
renal	0.40504147
neurological	0.40088261
vaccine	-0.33754695
pneumopathy	0.23394798
oxygensat	0.20272572
respdistress	0.17682461
asthma	-0.16616067
dyspnoea	0.15002913
cough	-0.12900262
diseaseCount	0.10675833
sex	0.10148033
diabetes	0.08526530
diarrhea	-0.06120880
fever	-0.04505276
obesity	0.04457232
sorethroat	-0.03844308
vomit	-0.03331888
age	0.03074600
cardio	-0.03031399
icuStayDuration	0.01593533
symptomsCount	0.01084671
hospStayDuration	-0.00577876
hematologic	0.00000000

This is the output for the Lasso model, however, it had high error rates and we found it to not be sufficient since we found that Elastic Net performed even better. This model showed that ICU had the greatest impact, even more than age which would have been an interesting discussion if this was chosen to be the best model.

Appendix 3



This residuals plot shows that although this model is largely linear, it defies the linearity assumption where the residuals are normally distributed.

Appendix 4

Elastic Net – The coefficient for age became significantly higher when introducing a nonlinear component to it, the coefficients were highest when using a spline component for age with degrees of freedom 2, this is true for part 2 as well. This shows that age is definitely not a linear predictor.

Polynomial with Elastic Net

	Coefficient
poly(age, 2)1	45.799254840
poly(age, 2)2	8.663733468
icu	0.376165214
(Intercept)	0.167678064
vaccine	-0.082638877
renal	0.057424517
immuno	0.054768760
neurological	0.048480227
oxygensat	0.035800265
diseaseCount	0.030744792
dyspnoea	0.026231016
cough	-0.021104341
respdistress	0.019424032
asthma	-0.016215016
sex	0.015362117
pneumopathy	0.008982062
cardio	-0.008401646
diarrhea	-0.002009015
fever	0.000000000
sorethroat	0.000000000
vomit	0.000000000
hematologic	0.000000000
downsyn	0.000000000
diabetes	0.000000000
obesity	0.000000000
hepatic	0.000000000
icuStayDuration	0.000000000
hospStayDuration	0.000000000
symptomsCount	0.000000000

This shows instability due to the extremely high polynomial age coefficient estimates which is a concern since it shows the model will not perform equally as well on unseen data and also had a high error rate.

Appendix 5

Comparison Table with all models in part 1

Model	Misclassification Rate	F1 Score	AUC
Lasso	0.191377	0.782541	0.7181733
Ridge	0.1808712	0.787778	0.784037
Net Elastic	0.1800661	0.792517	0.7963935
Polynomial with Net Elastic	0.1800541	0.792522	0.796398
Spline with Net Elastic	0.1800481	0.793619	0.7963995

This table shows that Lasso was clearly the weakest model.

Appendix 6

The spline with Elastic Net was the best choice as it minimised error for the cross-validation across the 10 folds and maximised performance on both metrics.

Part 2

Regularisation Models

To construct this model we used 10 fold cross-validation to determine what the best lambda values were (shrinking penalty coefficient). We constructed a grid of lambda values with a wide range and many breaks to really capture the best value. We then decided to use the value that was one standard deviation from the best to minimise the bias-variance trade-off which is prominent in regularisation models. We then tuned the threshold for acceptance and rejection and found that to be 0.3 for both models. We found this model to become too oversimplified since all the coefficients were shrunk (with 8 predictors being shrunk to 0 in Lasso) and since there are already very few predictors available, with the regularisation, not enough predictors are left which makes this model bad.

We found that these models performed poorly

Model	Misclassification Rate	F1 Score	AUC
Lasso	0.2866013	0.61055	0.6938214
Ridge	0.2917032	0.60998	0.688991

Appendix 7

This shows that the model had very poor predictive performance, hence we opted against this model.

KNN Model

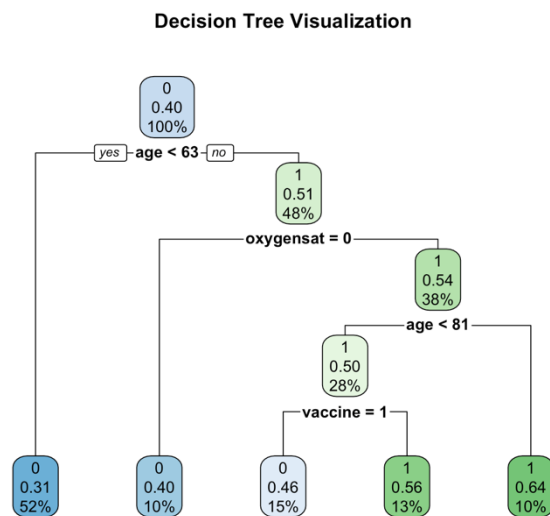
When fitting the KNN model, we must use cross-validation to determine the optimal k value, we tested this through various odd k values until the misclassification rate was minimised which was achieved with a k value of 5. When examining the F1 score and AUC values, we found them to be maximised with a k value of 5. KNN balances variance and bias well however, it was not a strong predictive model for us since the F1 score was only 0.60932. The AUC was also only 0.6569. This means that the model is weak at differentiating between cases but also the precision and recall of predictions.

Appendix 8

Single Decision Tree

We started with selecting a decision criteria, we opted for the Gini Impurity index since we favour larger partitions in each node, i.e., a purer node. A tree is made recursively, with each node splitting the data into subsets that are as pure (homogenous as possible). We then used cross-validation to evaluate the model's performance. When we were deciding the metric to build it on to maximise the performance, we chose F1 score for the reasons discussed in the report. We fit the tree using 'rpart'

which means that during training, it performs cross-validation to assess the performance of different hyperparameter values, hence we did not have to manually compute this.



We found this tree to perform very poorly with an F1 score of only 0.54542, this is because singular decision trees are very prone to overfitting and it does not capture many of the underlying trends since it is heavily oversimplified with limited predictors.

Appendix 9

Random Forest

When tuning a random forest there are a plethora of hyperparameters which we had to tune. First is the number of trees, although we found that it was most suitable to use half the number of predictors we had since we had a relatively smaller number of available predictors and using the recommended rule of square root of p would not have been appropriate, which is why we defined the `mtry` parameter as 8. We also found that the computation time was too high with 500 trees which is what we started with, hence we decreased it to 50 then continued to experiment with different tree sizes but finally settling on 50 to maintain a good balance between computation time and accuracy. We used the `train` function in R to use cross validation using the metric of F1 to improve results. We used a `tune` grid to tune all other hyperparameters to ensure we got the best possible tree when providing all the predictors known upon admission.

We then also generated a variable importance plot based on the Gini index. This was to be able to understand what were the most important factors in this random forest since this is much more of a qualitative option.

However, we found the F1 score to be quite low, only 0.59156, hence we opted against this model.

Appendix 10

Full Logistic Model

When fitting this model, we had to tune to find the best threshold, which was found to be 0.3. We also experimented with dropping different predictors that were not significant and found that dropping 2 predictors `vomit` and `fever` improved the F1 score to attain 0.61096 with the adjusted threshold. However, this was not enough hence we also looked into interaction terms based on their correlations and how well they decreased the cross validation misclassification error.

We also experimented with nonlinear terms as discussed in the report.

The interaction terms we experimented with were

- `fever_cough`
- `diabetes_obesity`
- `fever_cough_dyspnoea`

- fever_dyspnoea
- dyspnoea_cough
- fever_cough
- vaccine_sorethroat
- fever_cough_sorethroat
- hematologic_diarrhea
- fever_sorethroat
- fever_vaccine
- age_vaccine
- fever_diabetes
- vomit_hematologic
- asthma_dyspnoea
- diabetes_pneumopathy
- dyspnoea_vaccine
- oxygensat_obesity
- sex_dyspnoea_oxygensat_diabetes_neurological_pneumopathy_obesity_hematologic
- vaccine_diarrhea_cough
- sex_dyspnoea_oxygensat_diabetes_neurological_pneumopathy_obesity
- sorethroat_cough

Appendix 11

Final Model

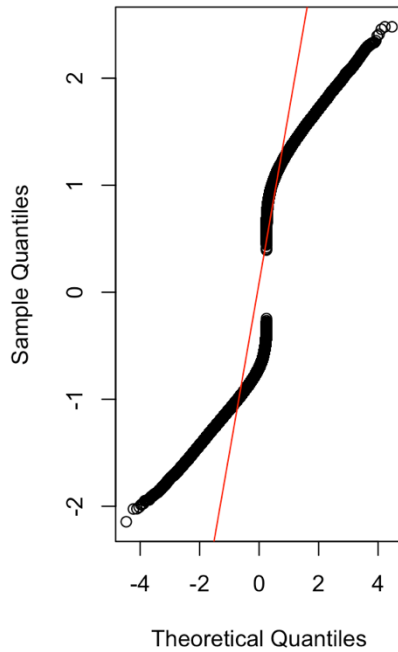
In our final model, we tuned the threshold and found it to be 0.305 which is quite similar to the rest of the thresholds and decided on including one interaction term of fever_vaccine and a natural spline for age. We found all the predictors to be significant at the 0.001% significance level. The analysis of the coefficients is in the report. The F1 score for this model was 0.61257 which was the highest F1 score between all models.

	GVIF	Df	GVIF^(1/(2*Df))
ns(age, df = 2)	1.392611	2	1.086319
sex	1.015170	1	1.007556
vaccine	1.837146	1	1.355413
cough	1.060458	1	1.029785
sorethroat	1.059386	1	1.029265
hematologic	1.004282	1	1.002139
downsyn	1.009933	1	1.004954
diarrhea	1.021190	1	1.010540
dyspnoea	1.115212	1	1.056036
oxygensat	1.103176	1	1.050322
asthma	1.014405	1	1.007177
diabetes	1.042116	1	1.020841
neurological	1.028979	1	1.014386
pneumopathy	1.018757	1	1.009335
obesity	1.103816	1	1.050626
fever_vaccine	1.630567	1	1.276936

To confirm our final model was viable, we examined the VIF values and found that they were all suitable since they were much less than 10, this portrayed that there is no multicollinearity concern.

Appendix 11

Q-Q Plot of Residuals



QQ plot of residuals of the final model emphasises that the model does not have a linear fit since the residuals are not normally distributed. This is a concern since some of the non-linear trends of the model may not have been captured.

This model is more deviated from linearity which should further analysed and addressed.

Appendix 12

Generative AI Appendix

- How do I remove high-leverage points from a dataset?
- How do I split my data into training and test data without the high leverage points?
- When do I convert my data into a matrix – for which machine learning concept?
- Can you create separate histograms for patients who died and those who were discharged?
- Can you overlay the histograms into one graph?
- Define a grid of lambda values for cross-validation.
- Perform cross-validation to find the best lambda for a ridge regression model.
- Can you provide the code for calculating AIC and BIC values and print the deviance value where it is one standard deviation from the mean?
- What is the error "number of observations in y not equal to the number of rows of x" and how do I fix it?
- Can you use the lambda one standard deviation away from the lowest deviance providing lambda?
- What parameters do I have to tune for lasso, ridge, logistic and tree models?
- What does cvglm.net do with logistic regression?
- How to make a QQ plot for the residuals of a plot?
- Why am I getting NA values in the output of my regression?
- How to fit a spline?
- What does the error mean where the size does not match, what could have gone wrong?
- What parameters do I tune for KNN models?
- How do I change the default of $m=\sqrt{p}$ to $\frac{1}{2}p$ for a random forest?
- When can I use cv.glm and cv.glmnet?
- How do I export coefficients of a regression into a table?
- How do I fit a KNN Model?
- how do I generate a variable importance plot in R?
- how do I code to find the cross-validation misclassification error?
- How do I calculate the F1 score and AUC in R?
- How do I use train method of cross-validation?
- How do I fit a random forest model?
- How do I fit a spline?
- How do I fit a polynomial?
- How do I export from R to csv?

Some examples of CHAT GPT's responses are below.

How do I calculate the F1 score and AUC in R?



To calculate the F1 score and AUC (Area Under the Curve) in R, you can use several packages, including `caret`, `ROCR`, and `pROC`. Below are steps and example code for each:

F1 Score

To calculate the F1 score, you can use the `caret` package, which provides a convenient function `F_meas`.

1. Install and load the `caret` package:

```
R Copy code  
  
install.packages("caret")  
library(caret)
```

2. Calculate the F1 score:

Assuming you have a vector of predicted labels (`pred`) and a vector of true labels (`truth`):

```
R Copy code  
  
pred <- factor(c("yes", "no", "yes", "no", "yes"))  
truth <- factor(c("yes", "yes", "no", "no", "yes"))  
  
confusion_matrix <- confusionMatrix(pred, truth)  
f1_score <- confusion_matrix$byClass["F1"]  
print(f1_score)
```

AUC

To calculate the AUC, you can use the `pROC` or `ROCR` packages. Below is an example using both.

1. Install and load the `pROC` package:

```
R Copy code  
  
install.packages("pROC")  
library(pROC)
```

2. Calculate the AUC using `pROC`:

```
R Copy code  
  
pred_prob <- c(0.9, 0.8, 0.4, 0.3, 0.6) # Predicted probabilities  
truth <- factor(c("yes", "yes", "no", "no", "yes"), levels = c("no", "yes"))  
  
roc_curve <- roc(truth, pred_prob)  
auc_value <- auc(roc_curve)  
print(auc_value)
```

3. Install and load the `ROCR` package:

```
R Copy code  
  
install.packages("ROCR")  
library(ROCR)
```

4. Calculate the AUC using `ROCR`:

Can you use the lambda one standard deviation away from the lowest deviance providing lambda?



Yes, you can select the lambda value one standard deviation away from the minimum lambda in the context of regularization techniques like Lasso or Ridge regression. This approach often helps in choosing a more regularized (and potentially more interpretable) model while still maintaining good predictive performance.

Here's how you can do this in R using the `glmnet` package:

1. Install and load the `glmnet` package:

```
R Copy code  
  
install.packages("glmnet")  
library(glmnet)
```

2. Fit the Lasso model and use cross-validation to find the optimal lambda:

```
R Copy code  
  
# Load necessary libraries  
install.packages("glmnet")  
library(glmnet)  
  
# Simulate some example data  
set.seed(123)  
x <- matrix(rnorm(100*20), 100, 20)  
y <- rnorm(100)  
  
# Fit the Lasso model using cross-validation  
cv_fit <- cv.glmnet(x, y, alpha = 1)  
  
# Extract the lambda values  
lambda_min <- cv_fit$lambda.min  
lambda_1se <- cv_fit$lambda.1se  
  
# Print the lambda values  
print(paste("Lambda with minimum deviance:", lambda_min))  
print(paste("Lambda one standard deviation from the minimum deviance:", lambda_1se))  
  
# Fit the model using lambda_1se  
final_model <- glmnet(x, y, alpha = 1, lambda = lambda_1se)  
  
# Display the final model coefficients  
print(coef(final_model))
```

Explanation:

1. **Load the `glmnet` package:** This package provides functions for fitting generalized linear models via penalized maximum likelihood.
2. **Simulate some example data:** This step is just for illustration. In practice, replace `x` and `y` with your actual data.
3. **Fit the Lasso model using cross-validation:** The `cv.glmnet` function performs cross-validation to find the optimal lambda values. Here, `alpha = 1` specifies the Lasso penalty (use `alpha = 0` for Ridge regression) ↓

How do I split my data into training and test data without the high leverage points



To split your data into training and test sets while removing high leverage points, you need to identify and exclude these points first. High leverage points are data points that have a significant influence on the fit of a regression model.

Here's how you can do this in R:

1. **Identify high leverage points:** You can use the `hatvalues` function to compute the leverage values and then determine a threshold to identify high leverage points.
2. **Remove high leverage points:** Exclude these points from your dataset.
3. **Split the remaining data into training and test sets:** Use functions like `createDataPartition` from the `caret` package to split the data.

Example:

```
R Copy code

# Load necessary libraries
install.packages("caret")
library(caret)

# Simulate some example data
set.seed(123)
n <- 100
p <- 5
x <- matrix(rnorm(n * p), n, p)
y <- rnorm(n)
data <- data.frame(y, x)

# Fit a simple linear model
model <- lm(y ~ ., data = data)

# Calculate leverage values
leverage_values <- hatvalues(model)

# Identify high leverage points (e.g., using a threshold of 2*(p+1)/n)
threshold <- 2 * (p + 1) / n
high_leverage_points <- which(leverage_values > threshold)

# Remove high leverage points from the data
clean_data <- data[-high_leverage_points, ]

# Split the data into training and test sets
set.seed(123) # for reproducibility
trainIndex <- createDataPartition(clean_data$y, p = 0.8, list = FALSE)
train_data <- clean_data[trainIndex, ]
test_data <- clean_data[-trainIndex, ]

# Display the number of high leverage points removed and the size of training and test sets
cat("Number of high leverage points removed:", length(high_leverage_points), "\n")
cat("Size of training set:", nrow(train_data), "\n")
cat("Size of test set:", nrow(test_data), "\n")
```

References

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